

Nhut Hung Nhan

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Summary

Detail-oriented Mathematics student with hands-on experience in computational math, numerical computing, programming, and scientific research.

Experience

Teaching Assistant, Carleton University, Ottawa

Sep 2023 - Current

- Completed coursework in Numerical Analysis, Ordinary Differential Equations, Mathematical Methods, Mathematical Reasoning, Calculus, and Linear Algebra.
- Recipient of the Mathematics and Statistics Teaching Assistant Excellence Award for 2024–2025.

Research Assistant, Carleton University, Ottawa

May 2025 - Oct 2025

- Modeling and analyzing quantum wave dynamics in deformed 2D graphene using tight-binding model (Schrodinger base) and finite element approximation model (Dirac base).
- Designing and modifying the model of graphene with lattice deformation for numerical experiments.
- Simulating time-dependent wave propagation using Kwant/tkwant.
- Integrating finite element modeling using FEniCS to solve PDEs and validate deformation models.
- Enhancing results of a time-independent tight-binding solver to compute eigenvalues, band structures, and local density of states (LDOS) for spectral analysis.

Research Assistant, Carleton University, Ottawa

May 2024 - Sep 2024

- Utilized Physics-Informed Neural Networks (PINNs) with JAX in Python for solving algebraic fractional systems.
- Conducted numerical experiments in MATLAB, achieving measurable improvements in accuracy or efficiency.
- Enhanced code efficiency for improved computational accuracy and complexity.
- Reviewed and revised research papers.

Skills

Programming:

- Python, Java, SQL, C, C++.
- LaTeX, MATLAB, R, Maple, Julia, Sage, CGSuite.
- HTML, CSS, JavaScript.
- Excel, VBA, Macros.

Libraries:

- NumPy, Matplotlib, Pandas.
- JAX, Scikit-learn, PyTorch.
- FEniCS, Pybinding, Kwant.
- MPI.

Soft Skills: Analytical problem solving, Research rigor, Technical communication, Mentorship and instruction, Collaborative work, Attention to detail, Self-directed learning.

OS & Tools: Linux, Git, CLI.

Projects

- Analyzed student academic outcomes (Predict Students' Dropout and Academic Success dataset) using PCA, hierarchical clustering, and Random Forest models to identify early predictors of dropout and characterize heterogeneous dropout subgroups using R.
- Implemented Fuzzy C-Means clustering in R on high-dimensional Human Activity Recognition dataset, optimizing cluster selection via elbow method to analyze the impact of the fuzziness parameter on cluster interpretability, demonstrating clear separation between dynamic and stationary activities.
- Implemented and experimented multiple regression models (Linear Regression, KNN, Decision Trees, Random Forest, SVR, Gradient Boosting) using Scikit-learn to predict outcomes, and evaluated performance.
- Implemented a Convolutional Neural Network with NumPy to recognize handwritten digits (MNIST dataset).
- Analyzed token growth and hallucination risks in LLMs using Python and DeepInfra API.
- Developed advanced VBA macros to automate complex workflows featuring conditional branching.
- Implemented MPI-based Finite Difference Schemes in C++ and visualized outputs in MATLAB.
- Developed and maintained a website for the CU Math & Stat Society using HTML, CSS, JavaScript.

Education

Carleton University

Sep 2025 - Current

Master of Science: Mathematics.

Carleton University

Sep 2021 - Apr 2025

Bachelor of Mathematics Honours: Concentration in Applied Analysis.

- CGPA: 11.35/12 - GPA: 3.9/4 - 94.6%.

Publications

- Emmanuel Lorin, Siming Tian and **Howl Nhan**. Simple and accurate microlocal derivation of low energy Dirac operators for locally deformed graphene. *Physica D: Non-linear Phenomena*. In review.
- Emmanuel Lorin and **Howl Nhan**. Data-driven fractional algebraic system solver. *Mathematics and Computers in Simulation*. 236:170–182, 2025.